Deepak Singh

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Professional Summary

Python Developer with **3+ years of product-based company experience** designing and building scalable backend systems and integrating AI capabilities into production applications. Hands-on with **Python (FastAPI, Flask)**, **REST API design**, **microservices**, and **event-driven architecture**, with a strong grounding in **OOP, SOLID, and clean architecture**. Practical experience with **LLM integration, model inference pipelines, and vector-based retrieval**, plus deployment using **Docker** on **AWS and Azure**. Comfortable working cross-functionally with backend, AI, and DevOps teams to ship modular, maintainable, and well-tested code.

Technical Skills

Programming Languages: Python, TypeScript, JavaScript, SQL

Frameworks and Libraries: FastAPI, Flask, Angular

Backend Engineering: REST API Design, Microservices Architecture, Event-Driven Systems (Redis Pub/Sub), High-Concurrency Systems, Integration Layers

AI/ML Integration: LLM Integration, Model Inference Pipelines, Prompt Engineering, Structured Output Validation

Databases and Data Handling: MongoDB (NoSQL), SQL, Redis (Caching, Pub/Sub), JSON Data Processing, Pandas, NumPy

DevOps and Cloud: Docker, Containerization, CI/CD Pipelines, Git, AWS (EC2, S3), Microsoft Azure

Engineering Practices: OOP, SOLID Principles, Clean Architecture, Design Patterns, Code Reviews, Agile/Scrum

Professional Experience

Software Engineer — Oct 2022 – Present

Openstream.ai | AI and Backend Systems — Bengaluru, India

- Designed and developed **Python RESTful APIs** using **FastAPI and Flask** as integration layers between core backend services and AI components, sustaining **sub-200ms response times** under high concurrency.
- Built and integrated **LLM-based inference pipelines** into production systems, covering prompt engineering, tool calling, and structured output validation for real-time AI features.
- Designed modular **microservices** and event-driven workflows applying **SOLID principles and clean architecture**, improving maintainability, testability, and scalability across distributed services.
- Re-engineered **Redis Pub/Sub and caching layer** powering event-driven flows, reducing P95 latency by **40%** and improving throughput stability under peak load.
- Optimized **Docker-based deployment workflows**, reducing image sizes by **35%** and supporting **99.9% uptime SLA** on cloud-deployed services.
- Built and maintained **CI/CD pipelines** (Git-based) for automated build, test, and deployment of backend and AI services, collaborating with DevOps on monitoring and performance tuning.
- Participated in **code reviews, technical discussions, and architectural planning**, applying clean coding practices that improved debugging efficiency and reduced mean time to recovery (MTTR).

Projects

1. Insurance Document Extraction Platform | Python, FastAPI, LLMs, AWS services

- Built a **FastAPI microservice** that extracts structured JSON from insurance documents using self-hosted LLMs, automating document understanding workflows for downstream systems.
- Designed a **multi-stage data pipeline** (conversion, classification, extraction, post-processing) supporting PDF, Excel, and DOC inputs with modular, testable Python components.

- Developed a **provider-agnostic LLM integration layer** with sync/async APIs, request-level configuration, and concurrency control to support high-throughput inference.
- Implemented document classification and retrieval using **vector-based search** and prompt-driven extraction, improving accuracy and consistency of structured outputs.
- Containerized services with **Docker** and deployed to **AWS**, enabling scalable and reliable backend execution.

2. 3D AI Avatar Interaction System | Python, FastAPI, LLMs, WebRTC, AWS

- Engineered the AI backend for a real-time conversational LLM avatar system, integrating prompt engineering, structured output validation, and WebRTC for low-latency interactions.
- Designed and maintained high-throughput **Python REST APIs** handling **10,000+ messages/hour** with sub-150ms latency, built for horizontal scalability on **AWS**.
- Implemented LLM output validation and **RAG pipelines** that enforced structured outputs and improved response reliability by **30%**.
- Deployed AI workloads as **Dockerized microservices** optimized for low latency and high availability.

3. Real-Time Session Management Dashboard | Flask, Redis, Microservices, MongoDB, CI/CD

- Engineered a **Retrieval-Augmented Generation (RAG)** pipeline integrated into backend services to ground LLM responses in domain-specific data, reducing hallucinations by **40%**.
- Achieved **sub-100ms dashboard refresh rates** using Redis caching and **event-driven Pub/Sub** architecture across microservices.
- Implemented role-based access control (RBAC), strengthening data security and access governance across the platform.
- Reduced support ticket volume by **25%** through real-time monitoring and proactive issue detection.

Education

Integrated B.Tech + M.Tech | Artificial Intelligence and Robotics 2017 – 2022
Gautam Buddha University Greater Noida, India

Senior Secondary School Jul 2014 – May 2016
M.P.V.M. Ganga Gurukulam Prayagraj, India